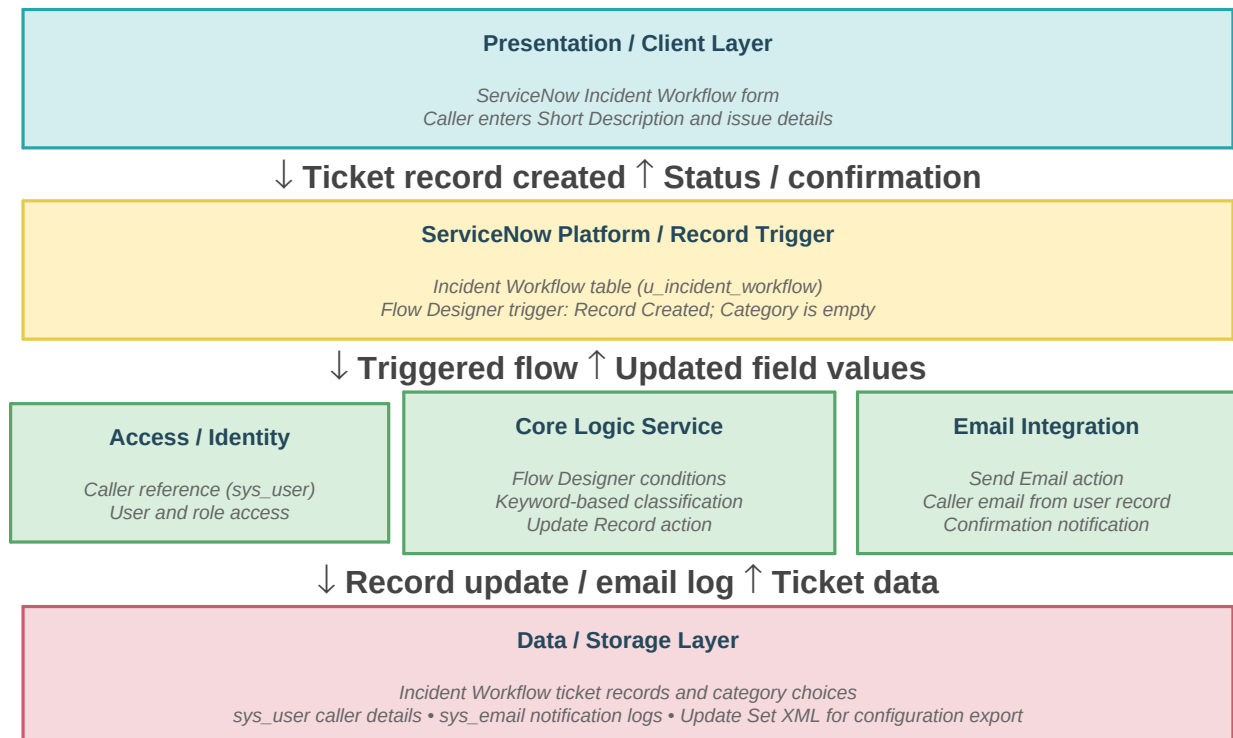


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-1038
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	5 Marks

Solution Architecture Diagram:

The architecture below shows how a caller submits an IT support ticket, ServiceNow Flow Designer classifies the issue, updates the ticket record, and triggers an acknowledgement email. The solution uses native ServiceNow platform components and configuration rather than a separate custom API gateway.



Component Description Table

The following table describes the role of each component in the Auto Ticket Classification workflow and identifies the ServiceNow technologies or configuration elements used.

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides the form where a student, teacher, or requester enters a short description and submits a ticket.	ServiceNow form; Incident Workflow table form
Record Trigger / Platform	Starts the automation when a new record is created and the Category field is empty.	ServiceNow Flow Designer; Record Created trigger
Core Logic Service	Checks description keywords and assigns the matching Category and Subcategory.	Flow Designer conditions; Update Record action
Dependent Choice Configuration	Restricts Subcategory choices based on the selected Category to improve data consistency.	ServiceNow Dictionary; dependent choice field
Access / Identity	Uses the caller reference to identify the requester and retrieve the caller's email address.	sys_user; ServiceNow roles and permissions
Email Integration	Sends a ticket submission acknowledgement and records the email activity for verification.	Flow Designer Send Email action; sys_email
Data / Storage Layer	Stores ticket fields, classification values, caller references, and notification records.	u_incident_workflow; sys_user; sys_email
Configuration / Deployment	Packages table, dictionary, and flow configuration for migration between ServiceNow instances.	Local Update Set; XML export

Classification Flow Summary

Keyword Trigger	Category	Subcategory
WiFi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Password / Forgot password	Access	Forgot Password
Slow Computer / Hanging	Performance	Slow computer

Architecture Notes

- The solution is designed around ServiceNow-native low-code/no-code components.
- No separate API gateway, load balancer, or external database is specified in the project documentation.
- Email delivery depends on the ServiceNow instance's email configuration and platform access.
- Keyword rules can be expanded with additional branches and assignment group routing in future versions.